

Can Non-Financial Signals Price Private Companies?

A Machine Learning Approach to Startup Valuation

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Problem

Private startups are hard to price: extreme information asymmetry, non-linear growth, and short histories make DCF impractical and comparables unstable during market swings. **Can non-financial signals substitute for traditional financial metrics in startup valuation?**

Data

- **3,403 venture-capital deals** from PitchBook, plus macro data from GEM (Global Entrepreneurship Monitor).
- **Out-of-time holdout** spanning the post-2022 correction: train on pre-April-2022 (bull, $n=2,723$), test on post-April-2022 (contraction, $n=680$) — a genuine regime shift for all models.

Method — a progressive 3-layer design

- **Layer 1:** OLS financial baselines (Revenue, EBITDA, Revenue Multiple).
- **Layer 2:** ML on *non-financial* features only (firmographics, deal context, investor syndicate).
- **Layer 3:** ML on the full set (Layer 2 + financials + macro) — an ablation to measure the marginal value of financial data.
- Models: Random Forest, Gradient Boosting, ElasticNet. Interpretability via SHAP.

Results (out-of-sample holdout, $n=680$)

Model	R ²	MAE
OLS Revenue Multiple (best financial baseline)	0.643	0.895
Random Forest — Layer 2 (non-financial only)	0.557	0.888
Gradient Boosting — Layer 2 (non-financial only)	0.558	0.897
Gradient Boosting — Layer 3 (full set)	0.552	0.901

- **Non-financial ML is competitive with the best financial baseline** — the Layer-2 Random Forest achieves the best absolute error (MAE 0.888).
- The OLS Revenue Multiple shows a higher R² (0.643) but the highest cross-validation instability, generalising poorly across regimes; the ML models are more robust.
- **"Information saturation":** adding financial and macro data (Layer 3) yields virtually identical performance to non-financial-only (Layer 2) — trailing financials add minimal value.
- **SHAP:** the collective capital capacity of the investor syndicate (mean AUM, ~31% of explained importance), not firm financials, is the dominant pricing driver — a certification effect.

Contribution

- Evidence that non-financial signals alone can price private companies competitively — and more robustly across market regimes — than financial baselines.
- A progressive, interpretable 3-layer framework that quantifies the marginal value of financial data (here, largely redundant).
- A foundation for the deeper question: which information is worth acquiring to support an investment decision under uncertainty.

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